

# Does adaptation compose?

## Independently trained language and genre modules on a low-resource intersection domain

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### Abstract

Domains that matter for the humanities sit at intersections of language, period and genre where almost no text exists, while the margins that contain them are data-rich. We ask whether adaptation to such a cell can be assembled from modules trained independently on those margins, and test it for a language factor and a genre factor on ⟨Swedish, literary⟩. The design forces a confound we make central: to keep the target unseen, neither module can be trained on its factor alone. The consequence is stark — neither module helps the target by itself, and their plain sum is worse than doing nothing — and yet averaging them helps, with no target data at all. What governs the composition turns out to be the *magnitude* of the two updates rather than their *direction*: three probes across two different spaces recover almost nothing, while one fitted scalar per module does the work, which suggests that interference-resolution merging addresses a problem this regime does not have.

### 1 Introduction

A concrete text domain is rarely defined by a single property. A 17th-century Dutch pamphlet and a contemporary Icelandic court ruling differ in language, in period and in genre simultaneously. Suppose a set of natural languages  $L$ , a set of periods  $P$  and a set of genres  $G$ : a domain is a cell  $\langle l, p, g \rangle$  of the grid  $|L| \times |P| \times |G|$ , and the cells that matter for the humanities are usually close to empty, while the marginals that contain them are not. Those marginals are what one can actually train on. This is the situation the ELMA project addresses: if the adaptation of a language model to a cell could be decomposed into contributions from each dimension, one could learn each contribution where data is plentiful and assemble a cell that was never observed.

Whether the factors of a humanities domain compose at all is an open question, and they may not: the variation of an intersection domain may have an irreducible structure of its own, not recoverable from its margins. We therefore organise the study around explaining an outcome rather than around demonstrating a method: we adapt a multilingual backbone independently to a language factor and to a genre factor, compose the two resulting modules on the intersection cell, and measure. The cell reported here is ⟨Swedish, literary⟩.<sup>1</sup>

**Neither module can be trained on its factor alone.** This is forced by the design. To keep the target cell unseen, the genre module can only be trained on the target genre in *other* languages, and the language module on the tar-

get language in the *other* genre. Here the “language” module is trained on legal Swedish and therefore encodes *legal* Swedish; the “genre” module is trained on literary text in fifteen languages none of which is Swedish, and therefore encodes literary *non-Swedish*.

Composing is therefore not adding two clean directions; it is adding two mixtures and hoping the unwanted halves cancel. It also makes a negative result ambiguous by default — the factors may fail to compose, or the confounds may compose badly — and separating those two readings is a design problem. Section 8.2 measures the confound directly.

### 2 Background and setting

We use two datasets, both parallel corpora distributed through the HuggingFace Hub and organised as language pairs.

`Helsinki-NLP/opus_dgt` is the legal half: the Translation Memory of the European Commission’s Directorate-General for Translation, i.e. EU legislation drafted once and translated into the official languages, and therefore formulaic, templatic, and built from the same source documents in every language. The copy on the Hub covers 10 languages, not the 25 the dataset card declares. `Helsinki-NLP/opus_books` is the literary half: out-of-copyright novels and their translations, four times smaller, far more varied in style and far more unevenly spread across languages. As published on the Hub, `opus_books` covers 16 languages and 49.0M words, of which 0.20M are Swedish, and `opus_dgt` 10 languages and 195.0M words, of which 11.14M are Swedish; words are whitespace-separated and counted over every pair containing the language, so repetitions across pairs are counted. *Genre* therefore takes exactly two values here, and the period dimension of the general ELMA framing is not exercised.

**The admissible grid is forced, not chosen.** A cell  $\langle l, g \rangle$  is usable only if  $l$  appears in *both* datasets: the genre module needs the other languages within the target genre, and the language module needs the target language within the other genre. The intersection of the two language inventories is four languages — Swedish, Finnish, Dutch, Spanish — giving  $4 \times 2 = 8$  admissible cells. Every design decision below is a choice among eight, not among four hundred.

**Why Swedish.** Of the four, Swedish fits the project best, for two reasons.

First, the target cell is scarce: the entire Swedish literary portion is 199,605 words, 0.41% of the literary corpus, against 3,657,410 for Dutch and 7,008,693 for Spanish. That single slice is the target cell, training, validation and test together. One qualification: it is the smallest of the four by raw count, but the ranking against Finnish (254,346) re-

<sup>1</sup>Code, the notebook with its saved outputs, the full training configuration and the corpus measurements: <https://github.com/mlahozy21/Modular-and-Compositional-Adaptation-of-a-Multilingual-LLM>

verses once repetition is removed. Each slice is one work in several translation pairs — three for Swedish, six for Finnish — and on exact-deduplicated text the Swedish slice is 75,763 words against 67,703 Finnish. Exact deduplication is a lower bound on the repetition, since differing segmentations leave near-copies intact. The scarcity argument holds against Dutch and Spanish by more than an order of magnitude; against Finnish it does not, and the case for this cell rests on the reason that follows.

Second, the modules can be trained on comparable, if not equal, amounts: 11.14M words of legal Swedish against 48.60M of literary non-Swedish, a ratio of 4.4 where the Finnish cell gives 2.85M against 48.43M, a ratio of 17. That reduces the imbalance rather than removing it: a small contribution from the language module is much harder to dismiss as starvation here than there.

**The target corpus, and how it is split.** The Swedish literary portion of `opus_books` is a single work appearing in three translation pairs. We split by chapter, contiguously, sending all three copies of a chapter to the same partition: the cover and chapters 1–12 to train (5,849 segments, 128,246 words), 13–14 to validation (1,184 / 25,929) and 15–19 to test (2,062 / 45,430).

### 3 Backbone and modules

#### 3.1 Backbone

The backbone has to make the question we are asking answerable, so two requirements disqualify outright and the properties that decide among what survives are measured on these corpora, not taken from model cards. We select BSC-LT/salamandra-2b [1], a 2.25B multilingual decoder trained on 35 European languages.

**Two requirements that disqualify.** First, Swedish must be in pretraining: if the target language is absent the language module *learns* Swedish rather than *adapting* to it, which is a different experiment with a different null result. It is a validity condition, not a preference. Second, the backbone must cover the fifteen languages of the genre module, or “genre” becomes entangled with “unseen language” inside a factor that is already confounded. Salamandra covers 14 of 15; the missing one is Esperanto, 0.42% of the module corpus.

**What decides.** Salamandra is the only candidate publishing a per-language word count, which lets us *state* how much Swedish the model saw instead of assuming it (EuroLLM publishes sources without counts; Qwen and Gemma publish neither). Swedish is 0.492% of the pretraining corpus, 18th of the 35 languages, against 1.173% for Dutch and 16.570% for Spanish; only Finnish among the admissible targets is rarer, at 0.341%. Fertility points the same way: measured on this corpus Salamandra needs 1.414 tokens per word on Swedish literary text against 1.593 for EuroLLM-1.7B [5] and 1.868 for Qwen3 [7], a 32% gap.

#### 3.2 Modular architecture

We use LoRA [2]: for each adapted matrix  $W$ ,  $W$  is frozen and a low-rank update  $\Delta W = (\alpha/r)BA$  is learned, with  $B$

initialised to zero so that training starts exactly at the backbone. A module is thus fully described by one displacement  $\Delta W$  per matrix. We adapt all seven projection matrices of every block ( $q$ ,  $k$ ,  $v$ ,  $o$ , gate, up, down) with  $r = 16$  and  $\alpha = 2r = 14,917,632$  trainable parameters. That is 0.66% of the model’s 2.25B, but the more informative figure is 1.24% of the 1.20B that sit in the transformer blocks: Salamandra’s 256k vocabulary puts the remaining 1.05B, 47% of the model, in the input and output embedding matrices, which we do not adapt.

#### 3.3 The two modules

Each module is trained on the whole of its corpus, one pass, with no data shared between them and none from the target: 696,334 segments and 11,135,049 words of legal Swedish for  $\tau_L$  (2,520 optimisation steps), and 2,483,074 segments and 48,601,840 words of literary non-Swedish for  $\tau_G$  (9,810 steps). Both learn what they were given (Table 1), which is the premise everything below rests on: a module that had learned nothing would make the composition question empty.

| module on its own held-out              | bpb <sub>0</sub> | bpb    | $\Delta$ | ppl         |
|-----------------------------------------|------------------|--------|----------|-------------|
| $\tau_L$ on legal Swedish               | 0.6991           | 0.5942 | −15.0%   | 7.11→5.30   |
| $\tau_G$ on literary $\mathcal{L}_{sv}$ | 1.0363           | 0.9048 | −12.7%   | 15.71→11.08 |

**Table 1:** Both modules learn their own factor. 14.9M trainable parameters each, one pass over the full corpus. Held-out here means held-out *blocks* of the same corpus, not held-out documents, so both figures are within-document generalisation and both are optimistic.

### 4 Composition

Two families of operator are available here, and the difference between them is what this paper ends up measuring. One acts on *how much* the composed update moves the model — the plain sum, the uniform average — and never looks inside the two displacements. The other acts on *where* it moves it, using the geometry of the updates to remove what they share. Both are built from the modules alone, so both are admissible where no target data exists at all. We build two operators of the first kind and one of the second, and let the target cell decide between them.

#### 4.1 Composing on magnitude alone

The two operators that need nothing but the modules themselves are the plain sum  $\Delta W_L + \Delta W_G$  and the uniform average  $\frac{1}{2}(\Delta W_L + \Delta W_G)$ . Neither has a parameter to fit. They differ only in total displacement, by a factor of two, which makes the pair a clean test of whether the composition is limited by how far it moves the model rather than by what the two modules disagree about. If the sum overshoots and halving it repairs the damage, the natural suspicion is that the two modules are writing into shared directions. The next subsection measures whether they are.

#### 4.2 What the two modules share

For each adapted matrix we take orthonormal bases of the two modules’ column spaces — the output directions

each one writes into — and compute the singular values of  $U_L^\top U_G$ , the cosines of the principal angles between them. The control matters here: two *random* rank- $r$  subspaces of the same dimension already share about  $\sqrt{r/d}$  of a cosine, so we report everything as a multiple of that chance level rather than as a raw overlap.

On this cell the overlap concentrates in `q_proj` and `k_proj`, at  $1.89\times$  chance, with the other five within a tenth of chance; and the fraction of  $\|\Delta W_G\|$  the projection removes is ordered the same way (Table 2), so the operator applies itself where the modules overlap without being told where that is.

| projection             | mean cos | chance | $\times$ chance | max    | removed |
|------------------------|----------|--------|-----------------|--------|---------|
| <code>q_proj</code>    | 0.1462   | 0.0772 | $1.89\times$    | 0.7384 | 3.8%    |
| <code>k_proj</code>    | 0.1460   | 0.0772 | $1.89\times$    | 0.7896 | 3.0%    |
| <code>v_proj</code>    | 0.0861   | 0.0772 | $1.12\times$    | 0.2536 | 0.5%    |
| <code>o_proj</code>    | 0.0811   | 0.0772 | $1.05\times$    | 0.2501 | 0.5%    |
| <code>gate_proj</code> | 0.0474   | 0.0449 | $1.06\times$    | 0.1142 | 0.2%    |
| <code>up_proj</code>   | 0.0474   | 0.0449 | $1.05\times$    | 0.1168 | 0.2%    |
| <code>down_proj</code> | 0.0847   | 0.0772 | $1.10\times$    | 0.2051 | 0.5%    |

**Table 2:** Principal angles between the two modules’ column spaces, by matrix type. Chance is the same quantity between two random rank-16 subspaces of the same dimension. The overlap concentrates in `q_proj` and `k_proj`, and the projection removes weight in the same order without being told where.

### 4.3 A composition operator with no free parameters

Where the two modules are orthogonal, summing is free: each writes directions the other does not touch. Where they are aligned, summing writes the same displacement twice, which is the mechanism behind the familiar observation that the best interpolation weight for merged models sits below one [8, 3]. Stacking instead of adding, as in MAD-X [6], is not open to us: it trains the second module on top of the first, and the task requires them independent. The standard response within weight space is to tune  $\lambda$ , but  $\lambda$  has to be fitted on target data, which the zero-shot setting forbids; it applies uniformly to matrices that do not need it; and it explains nothing.

We instead remove from the genre module the component the language module already writes:

$$\theta_0 + \Delta W_L + (I - U_L U_L^\top) \Delta W_G. \quad (1)$$

It has no free parameters, since  $U_L$  comes from weights we already have; it is self-targeting, because where the subspaces are orthogonal the projection is nearly the identity, so we never choose which matrices to apply it to; and it stays in the family, since  $(I - U_L U_L^\top) \Delta W_G = s(P^\perp B_G) A_G$  is still a rank- $r$  LoRA adapter, merged and served at identical cost. It is asymmetric — which module stays intact is a real choice — so we evaluate both orders. Because it acts on *where* the modules write rather than on how far they move the model, we return to it in Section 8.2 as the first of three probes into that question.

**It is falsifiable.** If the measured overlap is what makes the plain sum overshoot, this composition must beat the plain sum by a *margin that matters*. If it moves the score in the

right direction but negligibly, the overlap was epiphenomenal. We report either outcome.

## 5 Lightweight supervised adaptation

Besides the zero-shot setting above, the task allows adapting the composed model on at most 1M words of the target’s training split. The ceiling never binds here: the entire target cell is 199,605 words and its training split 128,246, so we use all of it. We spend that budget on the magnitude family of Section 4 again — now with the two scalars fitted rather than fixed at one or a half:  $\theta_0 + \lambda_L \Delta W_L + \lambda_G \Delta W_G$ . Two parameters, against the 14.9M of a fresh rank-16 adapter. The choice is not cosmetic: with 187k training tokens a fresh adapter has roughly 80 parameters per token, enough to learn the domain by itself and therefore enough to erase the very advantage the experiment is meant to detect.

Estimation is by exhaustive search on a grid of step 0.2 over  $[0, 1]^2$ , refined to 0.1 in the  $3 \times 3$  neighbourhood of the winner, scored on the *whole* training split (366 blocks, 753 kB). Validation and test are never consulted for the choice.

We also fit a per-layer variant,  $\lambda_L[\ell], \lambda_G[\ell]$  for each of the 24 blocks — 48 scalars, which does require gradient descent. It is initialised at the global optimum, so the two are nested and any gain is attributable to depth alone. The budget is fixed in advance rather than chosen by a stopping rule —  $\min(150, \max(30, 3N/16))$  steps for  $N$  training blocks, here 68 — and validation and test are read only once it has run out, so nothing in the fit can have consulted them. The  $\lambda$  are left unconstrained: since  $\tau_L$  hurts the target on its own, a layer that wants  $\lambda_L < 0$  is a result, and bounding them to  $[0, 1]$  would have hidden it.

## 6 Evaluation protocol

**Metric.** We report bits per byte —  $\text{NLL}/\ln 2$  over the UTF-8 bytes actually predicted — with perplexity alongside because that is what readers expect. Perplexity per token is not a property of a model on a text but of the model together with its tokeniser: a tokeniser that splits more produces easier tokens and a lower perplexity without predicting anything better, and fertility across the backbones we considered differs by 32%. Bits per byte normalises by the text itself, so it is invariant to that choice and still means something to a reader whose tokeniser is not ours. Within our tables one tokeniser is shared and the two metrics order identically: the choice matters for what the numbers mean, not for how the systems rank.

**Baselines.** In the zero-shot setting: the backbone  $\theta_0$ , each module on its own, and the three operators of Section 4 — sum, average, and projection in both orders. The supervised setting adds one more, and it is the one that decides what this paper may claim: each module alone *with its own scale fitted the same way*. Against an unscaled module any composition looks good, because part of what it buys is simply a smaller step; a composition that only matches a single *scaled* module has composed nothing.

**Statistical test.** Every system is evaluated on identical text, so the comparisons are paired and we bootstrap them with 10,000 resamples [4].

| system                              | bpb    | $\Delta$ | ppl   | 95% CI           |
|-------------------------------------|--------|----------|-------|------------------|
| $\theta_0$ backbone                 | 1.1045 | —        | 22.73 |                  |
| $\tau_L$ language only              | 1.1282 | +0.0237  | 24.31 | [+0.022, +0.026] |
| $\tau_G$ genre only                 | 1.1279 | +0.0234  | 24.29 | [+0.017, +0.027] |
| sum $\tau_L + \tau_G$               | 1.1424 | +0.0379  | 25.30 | [+0.032, +0.041] |
| mean $\frac{1}{2}(\tau_L + \tau_G)$ | 1.0942 | -0.0104  | 22.08 | [-0.015, -0.008] |
| proj $\tau_L + P^\perp \tau_G$      | 1.1411 | +0.0366  | 25.21 | [+0.031, +0.039] |
| proj $\tau_G + P^\perp \tau_L$      | 1.1415 | +0.0370  | 25.24 | [+0.032, +0.040] |

**Table 3:** Zero-shot composition on the target test split.  $\Delta$  is against the backbone; positive is worse. Intervals are the chapter-level paired bootstrap on the chapter-partitioned test (126 of 129 blocks).

|                                         | fitted | train  | test          |
|-----------------------------------------|--------|--------|---------------|
| $\theta_0$                              | 0      | 1.1369 | 1.1045        |
| mean (0.5, 0.5)                         | 0      | —      | 1.0942        |
| language only, scaled (0.2, 0)          | 1      | 1.1316 | 1.0993        |
| genre only, scaled (0, 0.4)             | 1      | 1.1249 | <b>1.0930</b> |
| grid ( $\lambda_L=0.3, \lambda_G=0.3$ ) | 2      | 1.1211 | <b>1.0888</b> |
| per-layer                               | 48     | —      | <b>1.0869</b> |

**Table 4:** Lightweight supervised adaptation. “fitted” counts parameters estimated on target data; every fitted row is selected on the training split and reported on test. The two single-module rows are the controls that matter: the gap between each of them and the two-scalar row is what the *other* module is worth once this one has been scaled. The two dashes are by design — the uniform average fits nothing, and the per-layer variant spends a step budget fixed in advance — so neither consults a training score.

*The resampling unit is the chapter.* The target is one work in three segmentations, so the blocks carrying a given chapter are three renderings of the same text and not three independent observations. Treating them as independent would inflate the sample from five chapters to 126 blocks; it narrows the intervals by a factor of about 1.6, on an independence that does not exist. We report only the chapter-level intervals.

*Five clusters bound what any interval can say.* Resampling five chapters with replacement admits only  $\binom{9}{5} = 126$  distinct outcomes — a coincidence of arithmetic with the block count above, and unrelated to it — so the percentile interval is, to a good approximation, the range of the five per-chapter differences. We therefore quote those five values wherever a contrast carries weight, and read the agreement of their signs as what actually supports it: five out of five in the same direction is  $p = 0.031$  under a one-sided sign test, the direction being predicted in advance rather than read off the data. That is the strongest statement five chapters can make.

*Two supports are in play, and we keep them apart.* The tables score systems on the full test split; every contrast and every interval is computed on the chapter-partitioned one, which drops the tail of each chapter. A difference quoted in the text and the same difference read off a table can therefore disagree in the fourth decimal.

## 7 Results

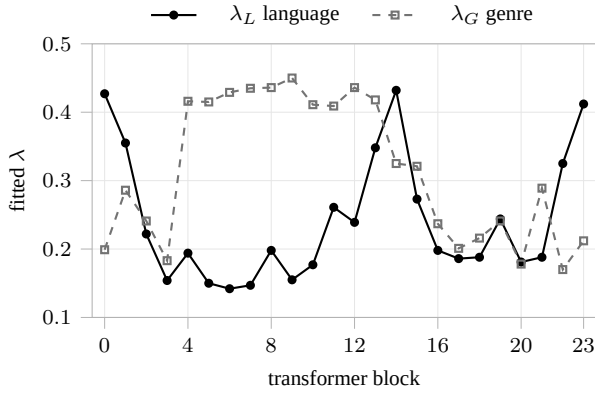
**Neither module helps, and the naive sum is much worse than nothing.** Table 3. Both modules *hurt* the target by almost exactly the same amount — +0.0237 for the language module trained on the target’s own language, +0.0234 for

the genre module trained on its own genre — and in every one of the five test chapters. This is the confound of Section 1 in its least convenient form: each module fails on the one cell it was built to serve. Typological distance does not look like the explanation for the genre module — Germanic languages are 40.8% of its training corpus and the target is Germanic — though more than half of that mass is English. Adding the two costs +0.0379, *less* than the sum of the parts (+0.0471): the two damages overlap rather than accumulate.

**But composition works once the scale is controlled.** Halving that sum turns +0.038 into -0.010: same direction, half the step. The mean beats the plain sum by -0.0486 ([-0.0504, -0.0473]) and improves on the backbone in all five chapters. Two modules that each damage the target, averaged, repair it — with no target data whatsoever. The overlap operator, meanwhile, behaves exactly as predicted and does not matter: it beats the plain sum in the predicted direction by -0.0014 ([-0.0018, -0.0009]) — real enough to exclude zero, small enough not to count, since halving the same sum is worth thirty-five times more. Order matters too, by 0.0004 bpb, with the language module the better one to leave intact. Section 4 named the epiphenomenal case — right direction, wrong magnitude — as the one that would refute the geometric account, and that is what we got.

**Most of the gain is one module at a fitted scale; the second is separable.** Table 4 is the comparison that decides what the paper may claim, and it is not the one usually made. Against the *unscaled* genre module the composition wins by -0.039 — but a genre module scaled the same way, one parameter fitted on the same split, already recovers -0.0118. Adding the language module on top is worth a further -0.0041, with the same sign in all five chapters (-0.0045, -0.0052, -0.0037, -0.0039, -0.0034) and an interval of [-0.0045, -0.0036] which, with five clusters, is essentially the range of those five numbers. The attribution depends on the order one decomposes in and we report both: crediting genre first leaves the language module worth 26% of the total, crediting language first leaves genre worth 67%. The two do not add to one because the modules are sub-additive — worth 0.0172 separately against 0.0159 together — so neither ordering alone is the answer. Averaging each module’s marginal contribution over both orderings, which for two players is the Shapley value, gives **30% to the language module and 70% to the genre one**. These are shares between two *mixtures*, not between a clean language factor and a clean genre factor: by Section 8.2 each module moves the target along its confound’s axis, so what the split divides is contaminated on both sides. Two scalars do not overfit: fitting them leaves the validation–test gap at +0.0327, no wider than the +0.0351 of the untouched backbone. We read nothing into the optimum’s symmetry — the training surface is flat to within 0.0001 bpb over  $\lambda \in [0.2, 0.4]^2$ .

**The two modules separate by depth.** The per-layer fit improves test by a further -0.0019, and the two profiles come out inverted (Figure 1): the language module is wanted at



**Figure 1:** The per-layer fit, initialised at the global optimum (0.3, 0.3). The language module is wanted at the edges of the network and the genre module in its middle; across depth the two profiles run opposite to each other, though with 24 points the correlation ( $r = -0.28$ ) does not by itself exclude zero. Forty-eight free parameters on one seed: the shape is reported as exploratory.

the two terminal blocks — adjacent to the embedding matrix and the output vocabulary — and the genre module in the interior. The surface properties of a language live next to the embeddings and the output vocabulary, and the interior is where the more abstract, style-like structure lives. The window is not one we chose: it falls out of a fit that was free to place  $\lambda_L$  anywhere. We report the profile as exploratory all the same, and the  $-0.0019$  as a number we would not defend alone.

## 8 Analysis

### 8.1 Where the gains and losses come from

Per-token NLL for each system on the test split, decomposed by position, token class, and corpus frequency — the last exactly, since the training blocks are still in memory and can be counted. Bytes are attributed token by token, which recovers 84% of the true count because decoding tokens in isolation loses leading spaces; that loss falls almost entirely on word-initial tokens, so comparisons across groups carry a bias we have not bounded.

**The gain lives at the cold start, and the damage where the corpus is thin.** The composition’s gain is sixteen times larger in the first eighth of a block ( $-0.068$  bpb) than in the last ( $-0.004$ ): with little left context the model falls back on its priors, and that is what the modules supply. The genre module shows the same thing from the other side, nearly harmless at the start ( $+0.010$ ) and steadily more damaging as context accumulates ( $+0.040$ ), which is what an over-scaled prior does once the evidence is in. On the frequency axis,  $\tau_L$  costs  $+0.005$  bpb on tokens its own corpus saw more than a thousand times and between  $+0.043$  and  $+0.082$  on everything else — up to seventeen times more. The rate is not monotone in rarity and most of the damage in absolute terms still sits in the mid-frequency bins, but the contrast between the top bin and the rest is large and one-directional: the confound is not a uniform fog.

**The loss is register-specific, and there is no division of labour.** The tokens  $\tau_L$  damages most are the comma and

the em dash, then *vi* (‘we’), the full stop, the semicolon, then *sade* (‘said’), *då*, *han*, *mig*. Administrative Swedish has no dialogue, no first person and no verb of speech, and the module trained on it degrades the backbone on precisely the tokens that mark narrative prose. What we do *not* find is the tidy story in which genre supplies lexis and language supplies morphology: both modules damage word-initial tokens by almost exactly the same amount ( $+0.025$  against  $+0.024$  bpb, on 78% of the bytes, that fraction taken from exact character spans rather than from the token-by-token attribution). And one caveat is large enough to state as a result: the newline byte alone accounts for **27% of the composition’s entire gain** in bits, eight times the next token. More than a quarter of our headline number is line-break placement, which is a property of how this corpus was segmented at least as much as it is genre. Setting it aside, what remains is function words and typography.

### 8.2 Three probes into where the modules act

The three probes are: project one module’s subspace out of the other in weight space, which is the operator of Section 4; measure the alignment of the two modules’ activation displacements; and remove from each module its component along the other factor’s axis, in state space. The middle one is a measurement; the other two are interventions, and are reported by what they buy.

**The modules are aligned in representation space, and the alignment cannot explain the scale.** The null here is not the one of Section 4: principal-angle cosines are singular values with chance at  $\sqrt{r/d}$ , whereas these carry a sign and average *zero* for independent pushes. Over 2,048 vectors per layer the mean cosine between  $\delta_L$  and  $\delta_G$  is  $+0.132$ , against a scale of  $1/\sqrt{d} = 0.022$  for a single random cosine — six times larger, and far outside what sampling noise could produce. So the two modules push the activations in substantially the same direction while writing into near-orthogonal weight coordinates. But a cosine of  $0.132$  predicts a per-module scale of  $1/(1+c) = 0.88$ , and the fitted optimum is (0.3, 0.3): over  $c \in [0, 1]$  the predicted scale never falls below one half, and reaching three tenths would need  $c = 2.33$ . Whatever forces  $\lambda$  down is not the double-counting the geometric account posits.

**The confound is a measurable direction — and it is the only thing the axes see.** We build two axes from backbone centroids with a single factor varying: genre as *sv literary* – *sv legal* (positive = more literary), language as *non-sv literary* – *sv literary* (positive = less Swedish). Decomposing each module’s displacement in that plane,  $\tau_L$  has a clearly negative genre coefficient ( $-0.084$ ) — it drags the target towards the *legal* pole, its confound — and  $\tau_G$  a clearly positive language coefficient ( $+0.048$ ), dragging it towards *non-Swedish*, its confound. But each module’s coefficient on its *own* factor’s axis is small or wrong-signed:  $\tau_G$  scores  $+0.001$  on the genre axis,  $\tau_L$  scores  $+0.030$  on the language axis. Within this plane, each module moves the target along the axis of what contaminates it and not along the axis it is named after. That is the claim of Section 1 in its strongest form, measured rather than inferred.

**The intervention that follows, and why it fails.** Removing that component — folded into the  $B$  factor of `o_proj` and `down_proj`, the two matrices writing into the residual stream, so the result is still a LoRA adapter — moves the test score by  $-0.0001$  bpb. The axes come from the target’s validation split, so this probe is not parameter-free and we compare it against the fitted  $\lambda$ , not the plain sum; we also fixed the variant to report in advance rather than choose among the four we ran, though choosing the best on test would not change the figure. The reason it fails is in the same decomposition: the plane accounts for 2.1% of  $\|\delta\|^2$  on average and at most 7.4%, so there is almost nothing to remove. A sharper probe would raise that fraction; what this establishes is that the correction available within the plane we could build is nil, not that no directional correction exists. For modules trained on factors this disjoint, the interference that TIES [9] and DARE [10] resolve appears not to be the binding constraint; total displacement is.

## 9 Discussion and limitations

Three things survive. First, composition on this cell is real but it is mostly a statement about magnitude: two modules that individually damage the target, averaged or scaled down, improve it, and one module at a fitted scale recovers roughly seven tenths of what the pair does. Task arithmetic already tunes a scaling coefficient [3]; what this cell adds is a measurement of what the second module buys on top of it —  $-0.0041$  bpb, consistent in sign in all five chapters. Second, direction is not where we could find any headroom: three corrections built on the geometry of the updates, in two different spaces, buy at most a tenth of what one scalar per module buys. Third, the confound of Section 1 is not a caveat here but the dominant effect, visible four times over: in both modules hurting their own target cell, in the language module’s damage concentrating on the punctuation and function words of dialogue, in that damage almost vanishing on the vocabulary its own corpus knows, and in each module displacing the target along its confound’s axis rather than its own.

**Limitations.** *Everything here is translated text:* DGT is a translation memory and `opus_books` novels with their translations, so the target and both modules are translations, and these two datasets cannot separate Swedish literary text from translated Swedish. *One work, three times:* the findings speak about one novel, with the consequences for sample size and for segmentation already given in Sections 6 and 8.1. *The modules are not compute-matched:* 48.6M words of genre against 11.1M of language, a factor of 4.4, so wherever the language module contributes little we cannot separate “this factor carries less” from “this module saw less text”. *No norm-matched control:* we claim total displacement governs the composition but never measured it, in state space, for the scaled single-module systems — one forward pass. *No direct-adaptation reference:* Section 5 argues only that such a run can obscure the effect, which is a reason not to substitute it for the composition and not a reason to omit it; as a normaliser we would reject it anyway, since 14.9M free parameters and an unconstrained recipe make that denominator a knob. *One seed:* the paired mea-

surement is precise because systems see identical text, but training variance is unbounded and is plausibly the larger of the two uncertainties here — the intervals we do report bound the smaller one.

## 10 Conclusion

Two modules trained independently, neither ever shown the target cell and neither of any use to it on its own, nevertheless model that cell better than the backbone once they are combined at the right scale — and they do it with no target data at all.

What decides that composition is how much the two updates move the model and not where they move it. We probed the direction three times, across two spaces, and recovered essentially nothing, while a single scalar per module did the work.

What the geometry does show, and show cleanly, is the confound the protocol forces on us: each module displaces the target along the axis of the factor it was contaminated with rather than its own. That is the finding we would carry to the next cell, because it is a property of the design and not of this one.

## References

- [1] A. Gonzalez-Agirre, M. Pàmies, J. Llop, et al. 2025. Salamandra Technical Report. arXiv:2502.08489.
- [2] E. J. Hu, Y. Shen, P. Wallis, et al. 2022. LoRA: Low-Rank Adaptation of Large Language Models. *ICLR*.
- [3] G. Ilharco, M. T. Ribeiro, M. Wortsman, et al. 2023. Editing models with task arithmetic. *ICLR*.
- [4] P. Koehn. 2004. Statistical Significance Tests for Machine Translation Evaluation. *EMNLP*.
- [5] P. H. Martins, P. Fernandes, J. Alves, et al. 2024. EuroLLM: Multilingual Language Models for Europe. arXiv:2409.16235.
- [6] J. Pfeiffer, I. Vulić, I. Gurevych, S. Ruder. 2020. MAD-X: An Adapter-Based Framework for Multi-Task Cross-Lingual Transfer. *EMNLP*.
- [7] Qwen Team. 2025. Qwen3 Technical Report. arXiv:2505.09388.
- [8] M. Wortsman, G. Ilharco, S. Y. Gadre, et al. 2022. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. *ICML*.
- [9] P. Yadav, D. Tam, L. Choshen, et al. 2023. TIES-Merging: Resolving Interference When Merging Models. *NeurIPS*.
- [10] L. Yu, B. Yu, H. Yu, et al. 2024. Language Models are Super Mario: Absorbing Abilities from Homologous Models as a Free Lunch. *ICML*.